

Respondent Covariates Can Lie: Evidence from Hierarchical Choice Models

An empirical stress test of model flexibility, new-person generalization, fitted heterogeneity, and decision consequences

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Abstract

A natural expectation when adding respondent covariates to hierarchical choice models is that useful information will matter while unhelpful information will mostly get out of the way. That expectation can fail.

We deliberately broke the respondent-level relationship between a binary characteristic and preference, then entered twenty shuffled versions of that characteristic as upper-level covariates. The ordinary specification looked extraordinarily successful in sample, receiving large rewards from both likelihood and the NR-LMD approximation.

On respondents held entirely outside estimation, the result reversed: prediction deteriorated sharply, while covariate-level regularization returned performance essentially to the no-covariate benchmark. Applied covariate cases produced smaller prediction differences. In a targeted case, better new-person prediction under regularization was not achieved by simply eliminating the differentiation the respondent information was added to capture.

The resulting modeling problem is one of attribution: how much systematic preference heterogeneity should a model assign to respondent information, and what evidence should be required to support it? The results establish a failure mode, not its prevalence, and evaluate one regularization response rather than a uniquely correct solution.

Model fit is treated as the beginning of the evidence chain, not the end. We follow the fitted structure through new-person transfer, heterogeneity attribution, and into quantities used for decisions.

Author note: This study is part of an ongoing research program applying the Minimum Evidence Standard to methodological claims in discrete choice modeling.

1. Introduction

Adding respondent covariates to a hierarchical choice model changes more than the number of predictors.

It gives the upper level additional ways to explain preference differences systematically. That can help when those relationships transfer to new people. An intuitive expectation is that respondent covariates carrying little useful information

will simply receive little posterior influence. If that expectation fails, the obvious remedy is stronger regularization. But that creates an analyst decision of its own: how much should the prior be tightened, for which covariates, and on what evidence?

In a controlled stress test, we added twenty respondent variables created by randomly shuffling an observed binary characteristic. The ordinary upper-level specification looked extraordinarily good in sample, with large gains in both likelihood and Newton–Raftery log marginal density. Then the result reversed on respondents held completely outside estimation. The same model predicted new respondents dramatically worse than the model without the added respondent covariates, while the regularized specification returned essentially to that benchmark.

That pattern is not confined to deliberately randomized covariates. With eight plausible respondent covariates, the regularized model again predicts new respondents better, although the difference in prediction is much smaller. The comparison using covariates based on brand consideration is the constructive case: with the same respondent information in both models, the regularized treatment predicts new respondents better without eliminating the targeted differentiation that motivated including those covariates.

Two labels will keep the distinction clear:

- **Permissive Covariates:** respondent covariates whose upper-level effects receive the model's ordinary prior scale simply because the variables are included.
- **Justified Covariates:** respondent covariates whose upper-level effects are estimated under a covariate-specific regularizing prior.

These labels describe two modeling conditions. They are not software packages or statements about how analysts generally work. Permissive refers to giving included respondent covariates the ordinary upper-level scale without covariate-specific evidence governing how much systematic differentiation they should support.

The statistical tool is group shrinkage. We use attribution to describe the question it addresses: how much systematic preference differentiation should the model carry through the respondent information? That is different from asking only whether a variable should be included. A respondent characteristic may contain useful information while still supporting much less systematic differentiation than the model can fit.

Prediction is only the first test. We also ask where the fitted heterogeneity goes and whether those differences reach quantities the model is being used to estimate. That takes us from new-person prediction to fitted heterogeneity, willingness to pay, and the revenue-maximizing price.

The stress test establishes that the failure can happen. It does not tell us how often it happens, and the two applied cases are not replications of the stress test. They ask a narrower question: whether the same directional comparison appears when the respondent information is substantively plausible or deliberately targeted.

That leads to a broader modeling principle: earned heterogeneity. Systematic preference differentiation attributed to respondent information should have to earn support from evidence appropriate to the claim being made.

The empirical contribution is the demonstrated failure mode and its consequences. The covariate-level shrinkage specification examined here is one response to that problem, not a claim that this regularization structure is uniquely correct.

2. Respondent Covariates and the Attribution Problem

2.1 The conventional hierarchical role

Start with a standard hierarchical choice model (Allenby & Rossi, 1998). The vector β_i contains the choice-model parameters for respondent i . Adding respondent covariates gives us the population model

$$\beta_i = z_i \Gamma + \xi_i, \xi_i \sim N(0, \Sigma).$$

The pieces have a straightforward interpretation. z_i contains observed information about respondent i . The term $z_i \Gamma$ moves that respondent's conditional population mean according to those characteristics. The remaining term, ξ_i , allows respondents with the same observed characteristics to differ from one another.

That is why respondent covariates can be useful. Without them, respondents share a common population mean while still differing through residual heterogeneity. With useful respondent information, the population mean can differ systematically across respondents. Rossi, McCulloch, and Allenby (1996), for example, use observable household characteristics to help inform preference estimates when relatively little individual purchase history is available.

There is nothing inherently problematic about this structure. The question is what happens when the relationship between respondent information and preferences is weak, unstable, or absent.

2.2 What previous work establishes

Previous work asks several different questions about respondent covariates.

One is how the relationship between respondent characteristics and preferences should be represented. Shively, Allenby, and Kohn (2000), for example, allow that relationship to be nonlinear. Another is whether respondent information improves fit or prediction. Chandukala, Edwards, and Allenby (2011) report improvements in both, using additional choices from respondents whose other choices contributed to estimation. Practitioners have also spent considerable effort asking which demographic, behavioral, and attitudinal covariates are useful in the upper level of HB models (Orme & Howell, 2009).

A different test is whether the learned relationship transfers to new people. Dotson, Büschken, and Allenby (2020) use validation respondents whose choice data play no role in estimating either their individual preference parameters or the population model. In one application, adding observed respondent covariates slightly improves estimation-sample fit while prediction for those new respondents becomes substantially worse.

What happens when the respondent-level relationship between the added information and preference has been deliberately broken?

2.3 Attribution rather than selection

Suppose a respondent characteristic contains some real information about preferences.

The question does not have to be whether to keep or discard it. A more useful question is how much systematic preference differentiation the model should attribute to that characteristic. We use attribution for that second question.

The concern is not heterogeneity itself. Large preference differences across respondents may be real and economically important. Once a respondent covariate enters the upper level, the model can place some of that heterogeneity into systematic relationships with the observed respondent information.

In-sample fit can reward those relationships, but whether they transfer to new respondents remains an empirical question.

2.4 Adding covariates creates prior opportunity

Adding respondent covariates changes the model before we know whether the added relationships will transfer.

Consider one preference parameter, k . Its systematic population component is

$$\mu_{ik} = \gamma_{0k} + \sum_{j=1}^J z_{ij} \gamma_{jk}.$$

Suppose the upper-level coefficients have a common independent prior,

$$\gamma_{0k}, \gamma_{1k}, \dots, \gamma_{Jk} \sim N(0, \sigma_\Gamma^2).$$

Conditional on respondent i 's covariate values, the prior variance of the systematic component is

$$\text{Var}(\mu_{ik} | z_i) = \sigma_\Gamma^2 \left(1 + \sum_{j=1}^J z_{ij}^2 \right).$$

The important part is the sum. Every additional covariate contributes another nonnegative term, giving the upper level another direction in which the conditional population mean can move. With several covariates entered at the same prior scale, those opportunities accumulate.

That is prior opportunity, not evidence of overfit. Whether the posterior uses that opportunity, and whether the fitted relationships transfer, are empirical questions.

Once respondent characteristics enter as columns of z , their substantive labels do not tell the model which relationships are better justified. Under a common prior scale, each covariate receives the same prior opportunity for systematic differentiation unless the analyst encodes some additional distinction.

3. Justified Covariates

Section 2 showed that adding respondent covariates creates additional opportunity for systematic preference differentiation. Justified Covariates change how much of that opportunity each covariate receives. The change is small: one covariate-specific scale is added to the upper-level coefficient vector.

3.1 The justification scale

The population model itself does not change:

$$\beta_i = z_i \Gamma + \xi_i.$$

For respondent covariate j , γ_j is the corresponding row of Γ . It contains that covariate's upper-level effects across the preference parameters. The difference between Permissive and Justified treatment is entirely in the scale given to that row.

	Permissive	Justified
Upper-level coefficient row	$\gamma_j = \sigma_\Gamma \tilde{\gamma}_j$	$\gamma_j = s_j \sigma_\Gamma \tilde{\gamma}_j$
Raw coefficient prior	$\tilde{\gamma}_j \sim N(0, I)$	$\tilde{\gamma}_j \sim N(0, I)$
Covariate-specific scale	$s_j = 1$	$s_j \sim \text{HalfNormal}(0, \sigma_s)$
Contribution to prior variance	$\sigma_\Gamma^2 z_{ij}^2$	$s_j^2 \sigma_\Gamma^2 z_{ij}^2$

Under Permissive treatment, every included covariate contributes at the ordinary upper-level scale. Under Justified treatment, that same contribution is multiplied by s_j^2 . Across all covariates, the systematic prior variance therefore changes from

$$\sigma_\Gamma^2 \sum_j z_{ij}^2$$

to

$$\sigma_\Gamma^2 \sum_j s_j^2 z_{ij}^2.$$

Each covariate still creates an opportunity for systematic differentiation. It no longer has to receive that opportunity at the ordinary scale. Putting the pieces together gives the full prior-opportunity calculation:

$$\text{Var}(\mu_{ik} | z_i, s) = \sigma_\Gamma^2 \left(1 + \sum_{j=1}^J z_{ij}^2 s_j^2 \right).$$

This is the same prior-opportunity calculation introduced in Section 2 with the covariate-specific scales added. When every $s_j = 1$, it reduces to the Permissive specification.

Under the Justified specification, $s_j \sim \text{HalfNormal}(0, \sigma_s)$.

The primary analysis uses $\sigma_s = 0.5$. We chose a scale below one to impose more prior skepticism on covariate-attributed differentiation than the Permissive specification, where $s_j = 1$. The value 0.5 is the scale of the prior, not a starting

value for s_j . The value was specified without reference to predictive performance in the validation sample.

The prior has no point mass at zero and does not prevent s_j from exceeding one. It therefore encourages contraction without forcing exclusion or imposing a fixed upper limit on the covariate effect.

The value of σ_s controls the strength of the regularization. It is a modeling choice, not part of the definition of Justified Covariates. We therefore refit the Twenty Placebos model with $\sigma_s=1$. The substantive result was essentially unchanged; details are reported in the appendix.

3.2 What gets regularized

The next question is what the justification scale actually shrinks.

For respondent i , the model separates preference variation into a systematic covariate component and a residual component:

$$\beta_i = z_i \Gamma + \xi_i,$$

The first term is the part carried systematically through observed respondent information. The second is residual respondent heterogeneity, with

$$\xi_i \sim N(0, \Sigma)$$

and

$$\Sigma = \text{diag}(\tau) \Omega \text{diag}(\tau).$$

Here, τ contains the residual standard deviations for the preference parameters, and Ω is their correlation matrix.

The justification scale acts directly on the first component. It does not directly tighten Σ , or the τ and Ω that define it. Because the components are estimated jointly, their posterior estimates can still change together. But the added regularization is placed specifically on the systematic relationship between respondent information and preferences.

Justified Covariates put the additional skepticism on the systematic covariate component.

Covariate-level shrinkage also differs from an include-or-exclude rule. A covariate can remain in the model while its full vector of systematic effects is strongly contracted. If the joint posterior supports larger effects through that covariate, those effects can remain despite the regularization. For Justified Covariates, the question is how much systematic differentiation survives regularization, not whether s_j crosses a threshold.

3.3 What the justification scale means

The justification scale is a component of the model. It is not a stand-alone measure of covariate quality.

The reason is visible in the parameterization:

$$\gamma_j = s_j \sigma_r \tilde{\gamma}_j.$$

The likelihood responds to the resulting upper-level effects. A larger s_j combined with smaller raw coefficients can produce effects similar to a smaller s_j combined with larger raw coefficients. The shared scale supplies structure across the coefficient vector, but s_j and $\tilde{\gamma}_j$ are estimated jointly.

The estimated s_j is therefore not the probability that a covariate is relevant. It is not an effect size or a direct measure of how much the covariate should be trusted. The substantive questions come downstream: how much systematic heterogeneity did the fitted covariate component produce, does that structure transfer to new respondents, and does it change willingness to pay, price response, or another quantity the model is being used to estimate?

The fitted consequences, not s_j by itself, tell us what the regularization accomplished.

3.4 What the mechanism establishes

The Justified specification provides a plausible mechanism for making poor-information covariates matter less.

If respondent information carries little useful signal, its systematic contribution can shrink hard while the covariate remains in the model. That is consistent with the expectation introduced earlier that irrelevant respondent information should receive little posterior influence. But the mechanism alone does not show that the fitted model actually behaves that way.

The next question is what evidence would be capable of establishing that claim.

4. Evidence Required for the Claim

A favorable statistic can establish only the claim that statistic is capable of supporting.

Better estimation-sample fit may show that a specification organizes the observed data well. It does not establish that the learned preference structure transfers to new people, that the result is internally coherent, or that the fitted model behaves credibly when used for substantive inference.

We use the Minimum Evidence Standard (Lee, 2025) to organize the evidentiary sequence. This paper does not test MES. The question here is whether respondent-covariate structure survives evidence appropriate to the claims being made: transfer to new respondents, internal coherence, and downstream behavioral or economic consequences.

Transfer is evaluated on respondents excluded from estimation. Internal coherence is checked for both the estimation-sample fit used in new-person prediction and the final full-sample refit.

Different claims require different evidence.

Analytic samples by evidentiary task		
Evidentiary task	Analysis sample	Purpose
In-sample fit and NR-LMD	n = 321 estimation respondents	Evaluates what the fitted models explain within the estimation data
New-person prediction	n = 321 estimation; n = 319 validation	Tests whether learned structure transfers to genuinely new respondents
Internal coherence	N = 640 full sample	Evaluates sampling and estimation behavior of the final fitted specifications
Fitted-heterogeneity attribution	N = 640 full sample	Describes where the final fitted hierarchy places heterogeneity
Behavioral and economic consequences	N = 640 full sample	Evaluates substantive implications of the final fitted specifications

Because the new-person prediction test is based on the 321-respondent estimation fit, we also report its corresponding diagnostics for completeness.

The fitted-heterogeneity decomposition uses the full-sample refits to show where the model places heterogeneity, and the downstream analyses use those same fits to examine the resulting behavioral and economic consequences.

4.1 Predictive evidence

The primary test of transfer is whether the systematic preference structure learned from the estimation respondents transfers to respondents whose choices played no role in fitting the model.

We evaluate that transfer using respondent-level posterior predictive log likelihood (Gneiting and Raftery, 2007). If a covariate specification improves estimation-sample evidence but predicts new respondents worse, the fitted covariate-conditioned structure has not demonstrated generalization.

4.2 Coherence evidence

A predictive result is difficult to interpret if the posterior was not sampled reliably.

We therefore examine chain agreement and standard sampler diagnostics (Vehtari et al., 2021). These diagnostics answer a limited question: can the result reasonably be explained by an obvious sampling or convergence failure? We examine rank-based

$\hat{R}$ and effective sample size (Vehtari et al., 2021), together with HMC diagnostics including divergent transitions, maximum treedepth, and E-BFMI (Betancourt, 2017).

Good computational diagnostics do not establish generalization. They make the predictive evidence interpretable.

4.3 Behavioral and economic evidence

The final question is what the fitted structure does when the model is used for the quantities the study is intended to estimate.

We examine willingness-to-pay differentiation, price-revenue relationships, and the revenue-maximizing price. These tests ask whether differences among the specifications propagate into quantities used for substantive or managerial inference.

Not every test needs to uncover a failure. A passed test is evidence too.

5. Empirical Setting and Experimental Design

5.1 Empirical setting

The empirical application is a commercial durable-goods choice study. The product category and brands are kept anonymous because their identities are not important to the methodological question. The choice model contains 28 product and price predictors. Respondents completed 14 dual-response choice tasks with four alternatives per task.

The setting is useful because the fitted model supports downstream quantities such as willingness to pay and price-revenue relationships, giving us a way to see whether differences in fitted heterogeneity reach quantities used for decisions.

5.2 Estimation and validation samples

The analysis uses a respondent-level estimation and validation split.

Of the 640 respondents, 321 are used for estimation and 319 for validation. The split is by person, not by choice task. None of the validation respondents' choice data are used to estimate their individual preference parameters or any population-level model parameters. Their choice data enter only when evaluating prediction for new respondents.

For validation respondent i , posterior predictive likelihood integrates over both uncertainty in the fitted population parameters and the respondent-level preference vector:

$$\log PPL_i = \log p(y_i^{\text{val}}, d_i | z_i, D_{\text{est}})$$

Where y_i and d_i are the first and second-stage responses and D_{est} is the estimation data.

The same draw of β_i is used across all 14 tasks for a respondent. We approximate this posterior predictive quantity by Monte Carlo averaging over posterior draws and draws from the new-respondent heterogeneity distribution, then take the log to obtain respondent-level log PPL.

After the new-respondent prediction test, each specification is refit to all 640 respondents for the fitted-heterogeneity and downstream analyses. These full-sample refits play no role in the validation results.

5.3 Benchmark and covariate regimes

The empirical comparisons use one no-covariate benchmark and three respondent-covariate regimes.

Regime	Added respondent information	Evidentiary role
Base	None	No-covariate benchmark
Twenty Placebos	20 shuffled binary respondent covariates	Deliberately demanding stress test of poor-information covariates
Feature Importance 8	8 binary top-box feature-importance measures	Plausible applied covariates of unknown usefulness
Brand Consideration 6	6 binary “definitely consider” measures tied to the corresponding brands	Constructive case: can regularization preserve the targeted differentiation that motivated inclusion?

Base contains none of the added respondent covariates and provides the benchmark for the comparisons that follow. Each of the other three regimes is estimated under both Permissive and Justified treatment, with the respondent information held fixed within the comparison. The difference is therefore how the same respondent information is regularized, not which information enters the model.

The Twenty Placebos regime requires additional description because the construction matters to its interpretation. Each placebo was created by randomly shuffling the binary respondent characteristic while preserving its prevalence in the split design; the twenty columns were generated once and frozen before fitting. All twenty placebo variables enter simultaneously, so this is one 20-covariate stress test rather than twenty independent replications.

For perspective, Dotson, Büschken, and Allenby (2020) use direct upper-level comparison models containing all 18 binary respondent measures in one application and all 53 in another.

Feature Importance 8 contains eight binary top-box measures of stated feature importance. These are plausible applied covariates whose usefulness is unknown, so this regime asks whether the Permissive–Justified comparison is confined to deliberately randomized respondent information. Brand Consideration 6 contains six binary measures indicating whether a respondent would definitely consider each corresponding brand; this is the targeted constructive case, asking whether regularization can improve transfer without simply eliminating the differentiation those covariates were included to capture.

The three covariate regimes therefore move from deliberately poor information, to plausible information of unknown value, to information deliberately targeted at the preference differentiation it is intended to explain.

6. Conventional Evidence Strongly Rewards the Permissive Model

The Twenty Placebos Permissive model looks extraordinarily good in sample.

Table 1 shows the central comparison. Higher values are better for both measures. The likelihood result is unsurprising in direction: adding upper-level flexibility gives the model more opportunity to fit the estimation data. The magnitude is harder to ignore. Mean log likelihood improves by about 1,907 units under Permissive relative to Base. Justified also improves slightly on Base on both measures, but we do not ask that small difference to carry any inferential weight here. The important contrast is the extraordinary in-sample reward for Permissive.

Table 1. In-sample evidence for the Twenty Placebos stress test

Specification	Mean log likelihood		NR-LMD	
	Estimate	Δ vs Base	Estimate	Δ vs Base
Base	−5,501	—	−5,658	—
Twenty Placebos — Permissive	−3,593	+1,907	−3,825	+1,833
Twenty Placebos — Justified	−5,457	+43	−5,632	+26

Note. NR-LMD denotes Newton-Raftery log marginal density; estimates use 1% trimming. Values are rounded to the nearest whole number. Higher values are more favorable.

6.1 The complexity objection

Raw likelihood naturally raises an objection. The Permissive model is more flexible, so better fit alone is not surprising.

We therefore asked whether an in-sample Bayesian model-evidence calculation that integrates over the prior would reverse the ranking. We used the Newton-Raftery approximation to log marginal density (Newton and Raftery, 1994) for the test of reversal. It did not.

The 1%-trimmed NR-LMD is approximately −3,825 for the Permissive model, compared with −5,658 for Base, a difference of about 1,833 log-density units. Justified remains near Base at approximately −5,632. Table 1 reports the 1%-trimmed calculation as a reference point rather than treating that trimming choice as uniquely correct. Across the examined range from 0% through 10% trimming, the Permissive model remains approximately 1,748 to 1,878 units above Base. The large separation and model ranking therefore do not depend on the particular 1% choice.

This comparison is not the central evidentiary test in the paper. Newton-Raftery provides a familiar in-sample model-evidence calculation, but its approximation is tail-sensitive and we do not treat it as definitive evidence that model complexity has been resolved. Other marginal-likelihood estimators, including Gelfand-Dey, can provide additional checks but are substantially more computationally demanding in this setting. We do not pursue increasingly costly marginal-likelihood calculations because they still address an in-sample model-evidence question.

Evidence of transfer requires prediction for genuinely new respondents.

6.2 The unresolved question

That apparent victory creates a problem rather than resolving one.

The placebo variables were created by randomly shuffling the respondent characteristic as described in Section 5. Chance relationships can still appear in the realized estimation sample, and the Permissive model is free to fit them. The in-sample evidence therefore cannot tell us whether the resulting preference structure will transfer to new respondents.

New respondents can.

7. New Respondents Reverse the Result

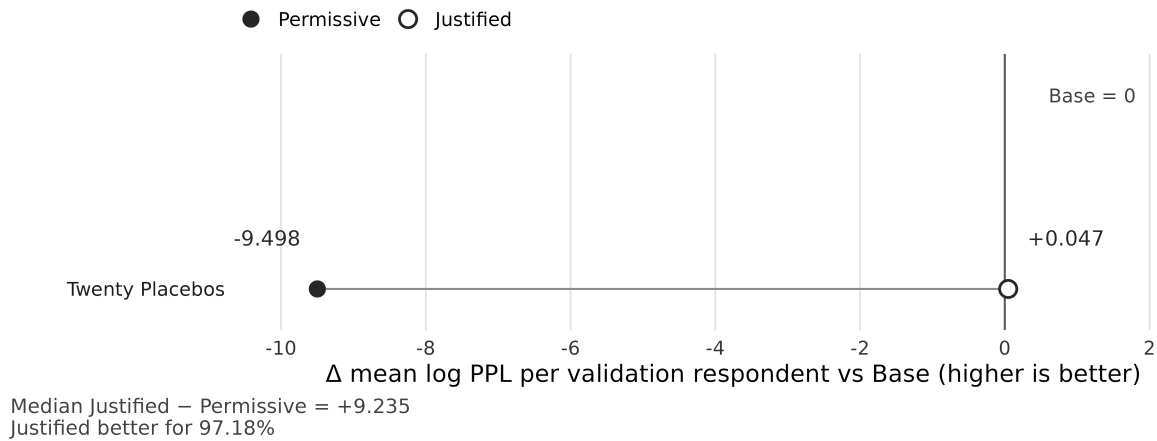
The apparent in-sample advantage does not transfer.

Figure 1 compares predictive performance for the 319 validation respondents. To make the comparison easier to read, each result is expressed relative to Base. Base has a mean log PPL of -23.311 per respondent and is shown as zero. Positive values indicate better prediction than Base; negative values indicate worse prediction.

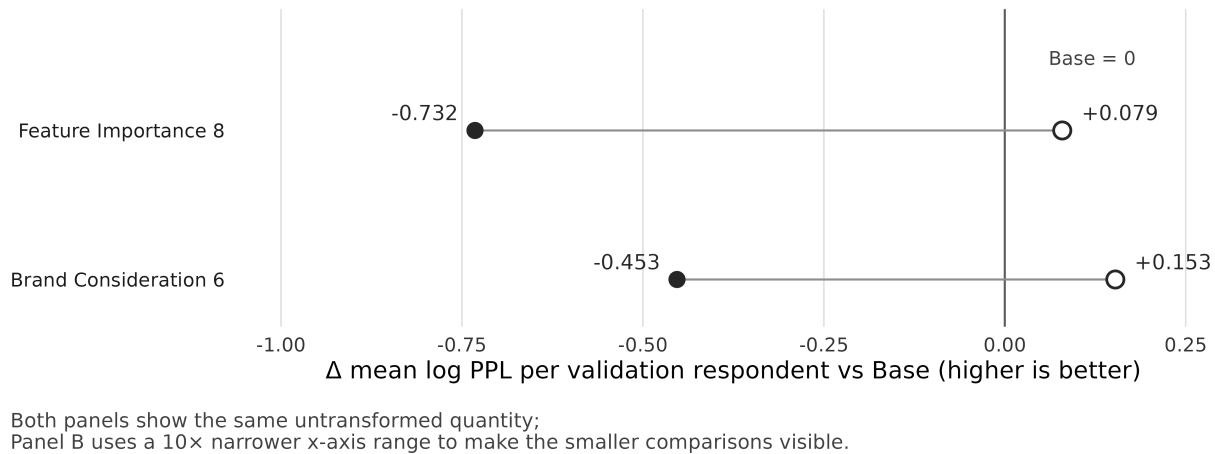
New-respondent predictive performance

Difference in mean log posterior predictive likelihood per validation respondent relative to Base

A. Twenty Placebos



B. Applied covariate regimes



7.1 Twenty Placebos

Start with the Twenty Placebos result.

Permissive falls 9.498 log PPL units per respondent below Base. In absolute terms, mean log PPL falls from -23.311 for Base to -32.809 for Permissive. The specification that looked extraordinarily good in sample performs dramatically worse on respondents whose choices played no role in estimation.

Justified produces a very different result. Its mean log PPL is -23.264 , only 0.047 above Base in this sample. The same twenty placebo covariates remain in the model; what changes is the regularization of their upper-level effects.

This is the essential stress-test result. Permissive respondent covariates can produce severe new-person overfit while receiving an enormous in-sample reward. In this stress test, covariate-level regularization returns predictive performance essentially to Base.

7.2 The mean is not hiding a few extreme respondents

A difference this large could still be misleading if it were driven by a small number of unusually bad predictions.

The result is not driven by a small number of extreme respondents.

For each validation respondent, we calculated the difference in log PPL between Justified and Permissive. The mean difference is +9.545 per respondent and the median is +9.235. Justified has higher log PPL than Permissive for 97.18% of the 319 validation respondents.

The median does one job here: it shows that the aggregate result is not being created by a handful of extreme respondents. We do not repeat this distributional check for every comparison below.

7.3 Feature Importance 8

Feature Importance 8 asks whether the direction of the Justified-Permissive comparison survives when the added respondent information is substantively plausible rather than deliberately randomized.

It does. The magnitude does not.

Permissive predicts 0.732 log PPL units per respondent below Base. Justified is 0.079 above Base, for a mean Justified-minus-Permissive difference of +0.811. Justified performs better than Permissive for 68.34% of validation respondents.

This is a much smaller result than the placebo stress test, and it should be treated that way. Its evidentiary role is narrower: in this application, the validation-sample advantage for Justified over Permissive is not confined to respondent information whose relationship with preference was deliberately destroyed.

It does not establish a general result about stated feature importance or about applied respondent covariates as a class.

7.4 Brand Consideration 6

Brand Consideration 6 provides the deliberately targeted constructive case.

Permissive predicts 0.453 log PPL units per respondent below Base. Justified predicts 0.153 above Base. Justified performs better than Permissive for 64.58% of validation respondents and better than Base for 57.99%.

The small improvement over Base should not carry more weight than it deserves. A slight predictive advantage over the no-covariate model is not, by itself, a reason to prefer the more complex specification. The constructive question is whether the

added respondent information can avoid harming transfer while still producing the targeted differentiation it was included to capture.

We return to that question later.

7.5 What the three comparisons establish

The three covariate regimes produce very different predictive differences. The realized validation-sample ordering favors Justified over Permissive in all three cases, but those comparisons do not carry the same claim.

The key distinction is that “Justified finishes 0.153 above Base” is not the constructive result. By itself, that tiny predictive advantage probably does not justify a more elaborate covariate model. The real constructive case requires the model to do two jobs at once:

1. not damage transfer; and
2. preserve the targeted differentiation that motivated adding the respondent information in the first place.

The Twenty Placebos stress test carries the strongest transfer result. Permissive fails dramatically on new respondents, while Justified returns essentially to Base. Feature Importance 8 shows the same directional comparison at a much smaller magnitude using plausible applied respondent information.

Brand Consideration 6 satisfies the first requirement in the realized validation sample, but Section 7 cannot establish the second. That requires evidence that the targeted differentiation survives and reaches the quantities the covariates were included to explain. A small predictive difference also does not establish that downstream consequences are small.

The predictive evidence establishes the transfer result. It does not finish the constructive case.

8. The Result Is Not an Obvious Computational Artifact

The predictive failure in Section 7 could have a simpler explanation: the Twenty Placebos Permissive model may simply have been estimated badly.

The available evidence does not point in that direction.

8.1 Coherence

For the Twenty Placebos Permissive model, the maximum $\hat{R}$ is 1.028. There are no divergences, no maximum-treedepth hits, and the minimum E-BFMI is 0.545 Betancourt (2017).

Sampling diagnostics for the Twenty Placebos — Permissive model				
Specification	Max R-hat	Divergences	Max-treedepth hits	Min E-BFMI
Twenty Placebos — Permissive	1.028	0	0	0.545
Note. Diagnostics refer to the fitted model used in the central Twenty Placebos comparison.				

These diagnostics do not establish that the model is substantively correct or that it generalizes. They answer a narrower question: whether the severe new-respondent failure can readily be attributed to an obvious HMC or convergence problem. The diagnostics provide no such explanation.

Because the new-respondent result in Section 7 comes from the 321-respondent estimation fit, we also report those diagnostics for completeness. They tell the same story: maximum $\hat{R}$ is 1.025, no divergences, no maximum-treedepth hits, and minimum E-BFMI is 0.600.

Sampling diagnostics for the Twenty Placebos — Permissive model				
Specification	Max R-hat	Divergences	Max-treedepth hits	Min E-BFMI
Twenty Placebos — Permissive	1.025	0	0	0.600
Note. Diagnostics refer to the $N = 321$ estimation-sample fit used in the Twenty Placebos new-respondent predictive comparison.				

Larger diagnostics for fitted models are reported in the appendix. Some other specifications contain isolated $\hat{R}$ or effective-sample-size issues, so the broader set of models should not be described as uniformly pristine.

8.2 Independent coding check

An independent coding check asks whether the unusual in-sample pattern depends on something peculiar to the primary analysis.

We repeated the random-covariate exercise using a separate code path and public choice data. Larger sets of random upper-level covariates again received substantial apparent in-sample rewards. Additional checks using a second random realization and a tighter fixed prior are reported in the appendix; they address reproducibility and prior sensitivity rather than software performance.

This exercise is a coding check only: it does not reproduce the new-respondent failure or test Justified Covariates.

9. How Much Heterogeneity Actually Passes Through the Covariates?

Section 2 established prior opportunity: adding respondent covariates gives the hierarchy additional latitude to create systematic preference differentiation. That calculation did not tell us how much of that opportunity the fitted model would actually use.

We can now measure it.

The question is not merely how much heterogeneity a model estimates, but where the model places it.

9.1 Decomposing fitted heterogeneity

To measure where the fitted heterogeneity goes, we separate each respondent-level coefficient into two parts.

For preference parameter k ,

$$\beta_{ik} = (z_i \Gamma)_k + \xi_{ik}.$$

The first is the systematic component associated with observed respondent information. The second is residual respondent heterogeneity. Under the fitted hierarchy, the residual component has mean zero conditional on respondent information and variance τ_k^2 .

For each posterior draw, define covariate-attributed heterogeneity as

$$H_k^{(C)} = \text{Var}_i \left[(z_i \Gamma)_k \right],$$

where the variance is evaluated over the empirical distribution of respondent covariates. Residual heterogeneity is

$$H_k^{(R)} = \tau_k^2$$

Total fitted heterogeneity is therefore

$$H_k^{(T)} = H_k^{(C)} + H_k^{(R)},$$

and the covariate-attributed share is

$$\frac{H_k^{(C)}}{H_k^{(T)}}.$$

This is a model-implied decomposition, not the empirical variance of respondents' posterior mean coefficients. For each posterior draw, the systematic component is evaluated jointly across the observed respondent covariate distribution, so correlations among the respondent covariates remain part of the calculation rather than being allocated separately to individual covariates.

9.2 From prior opportunity to posterior use

The prior and posterior quantities answer different questions.

Prior opportunity tells us how much systematic differentiation the specification makes possible before fitting the model. The posterior decomposition tells us how much systematic differentiation the fitted hierarchy actually carries through the respondent covariates. In the placebo stress test, that distinction matters because the respondent-level mapping between the shuffled characteristic and preference was deliberately broken.

Chance relationships can still appear in the realized sample, and the decomposition shows how much fitted heterogeneity the model ultimately routes through them.

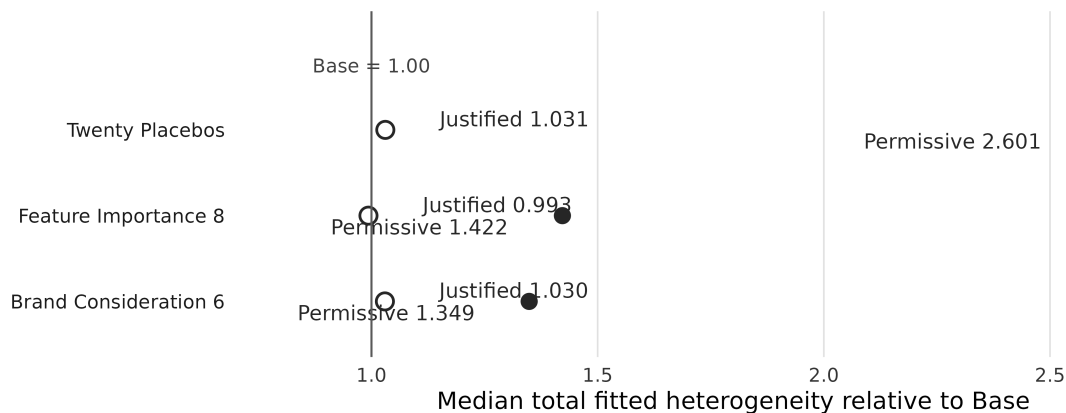
9.3 Permissive changes both how much and where

Figure 2 summarizes the full-sample refits in two ways. Panel A asks how much total heterogeneity the fitted model produces relative to Base. Panel B asks how much of that heterogeneity the model carries through the respondent covariates.

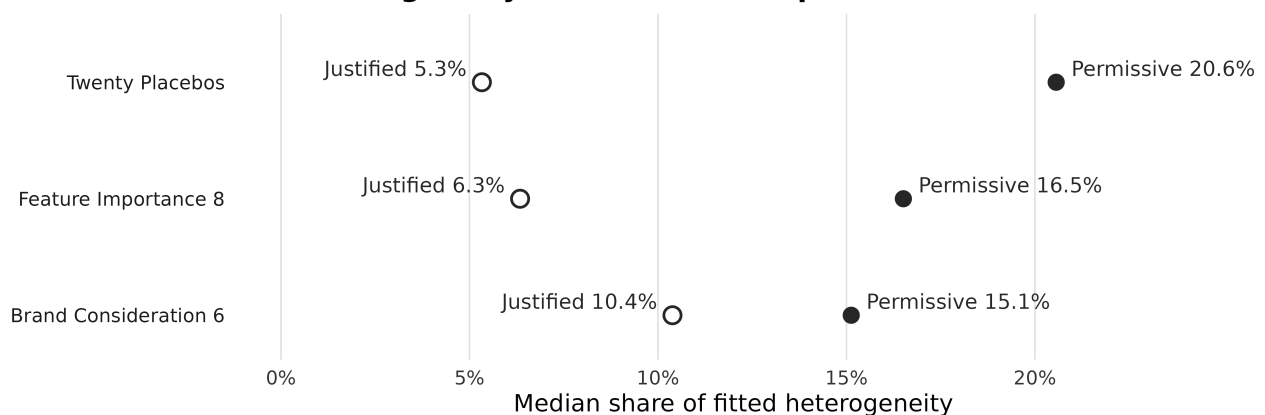
Fitted heterogeneity and its attribution

Median across 28 preference parameters

A. Total fitted heterogeneity relative to Base



B. Share of fitted heterogeneity attributed to respondent covariates



Base serves as the manuscript benchmark.

Panel B reports the share of fitted heterogeneity attributed to the respondent-covariate component under each specification

The contrast is not simply about where a fixed amount of heterogeneity is placed. In the specification that failed to transfer, Permissive fits substantially more total heterogeneity and routes a larger share of it through respondent information whose mapping to preference was deliberately broken. Under Justified, some fitted heterogeneity still passes through the covariates, but total heterogeneity remains close to Base. In this stress test, that is what the difference between the failed and successful transfer results looks like inside the fitted hierarchy.

The applied regimes are descriptive rather than separate inferential tests. For Feature Importance 8 and Brand Consideration 6, the differences are much smaller, and treating the close comparisons as ties would not change the argument. The useful pattern is that Justified remains near Base in total fitted heterogeneity while

still allowing some systematic differentiation to remain when the respondent information has substantive grounding. The fitted decomposition, rather than the justification scale itself, is the substantive result here.

Brand Consideration is useful for a different reason. Under Justified, the covariate component remains nontrivial rather than collapsing toward zero. That keeps the constructive case open rather than resolving it by simply eliminating the targeted structure.

The second requirement remains: does that surviving targeted differentiation reach the quantities the respondent information was added to explain?

Section 10 answers that question.

10. Do the Fitted Differences Reach Decision-Relevant Quantities?

Section 7 established the transfer result. Section 9 showed how the fitted models attribute heterogeneity between the respondent-covariate component and residual heterogeneity.

The remaining question is whether those differences in attribution reach quantities the model is being used to produce.

That question has two sides. The placebo case asks whether unsupported covariate-attributed differentiation can reach a decision-relevant output. The Brand Consideration case asks whether regularization can preserve the targeted differentiation that motivated adding the respondent information in the first place.

10.1 Unsupported structure can reach a pricing result

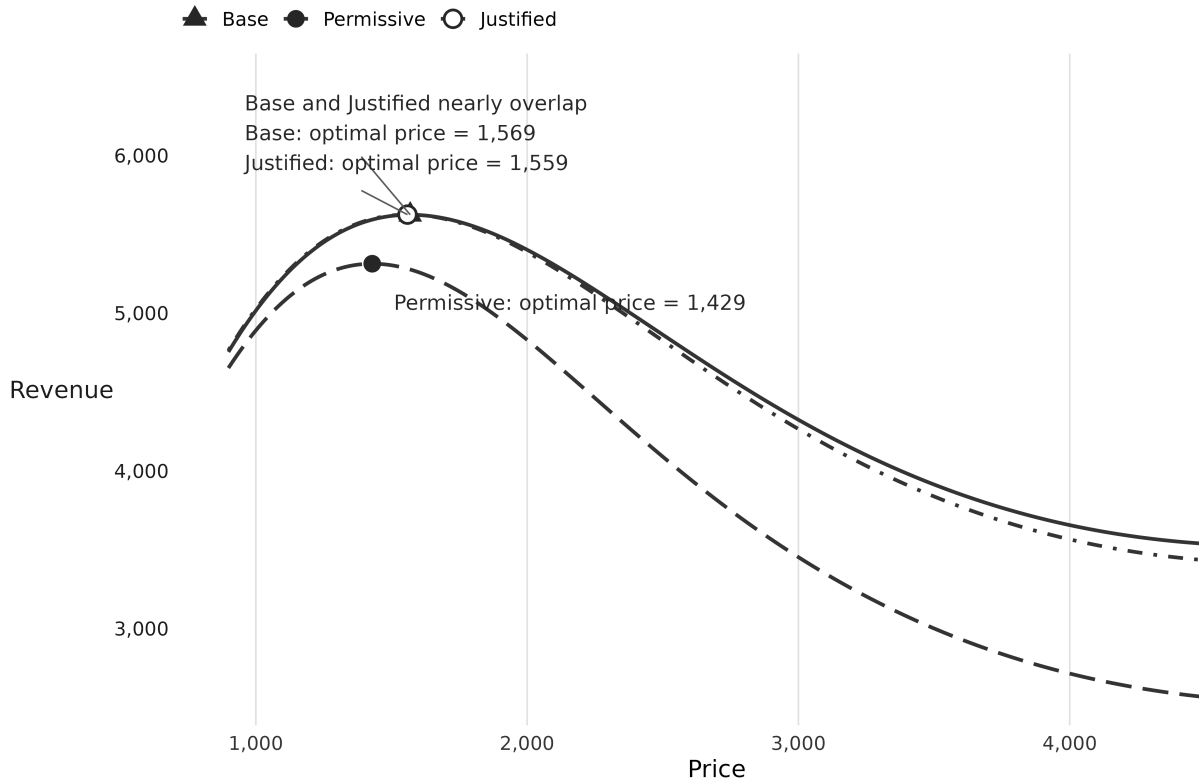
Placebo05 is an illustration, not a representative result.

It was selected because it provides a clear example of a downstream consequence, and its magnitude should not be interpreted as typical of the twenty placebo variables. The new-respondent failure was already established in Section 7 without relying on this example. The question here is narrower: can unsupported covariate-attributed differentiation reach a quantity of the kind ordinarily used for pricing decisions?

Figure 3A shows the complete price-revenue curves over the price range tested in the study for the subgroup defined by Placebo05. The revenue-maximizing price is approximately \$1,569 under Base, \$1,559 under Justified, and \$1,429 under Permissive. Justified is one \$10 grid step from Base, while Permissive moves the optimum downward by \$140.

A. Revenue response to price

Placebo 05 subgroup



Points mark the revenue-maximizing price on the tested price grid.

The curve matters as much as the location of the maximum. The result is not three isolated argmax values. Figure 3A shows that the fitted revenue relationship itself changes across the tested price range.

No manager would price a product using a deliberately randomized subgroup. That is not the claim. The result is that unsupported respondent-covariate structure can propagate far enough through the fitted model to alter the same kind of output that would be interpreted substantively if the covariate had ordinary business meaning.

10.2 Targeted brand differentiation remains under Justified

Brand Consideration asks the complementary question. Respondent covariates are included in hierarchical models partly because analysts want systematic differences among consumers that are useful for understanding choice. A regularization method that achieved better prediction only by eliminating those differences would therefore solve the problem in an uninteresting way.

Here willingness to pay is defined within a fixed competitive scenario as the change in the focal product's revenue-maximizing price when an attribute moves from a baseline level to a new level:

$$WTP_{jks}^{a \rightarrow b} = p_{js}^* (x_{jk} = b) - p_{js}^* (x_{jk} = a).$$

Here j indexes the focal product, k the attribute, s the competitive scenario, a the baseline level, and b the new level.

For these simulator comparisons, the quantity labeled WTP is based on the product's revenue-maximizing price within a fixed competitive scenario. For each Brand Consideration group, we use the corresponding brand as the reference and compare the subgroup with the total sample. This difference-in-differences asks how the target brand's position changes relative to each of the five competing brands, regardless of whether that change arises because the target brand becomes more attractive, the competitor becomes less attractive, or both. The target-brand contrast is therefore zero by construction.

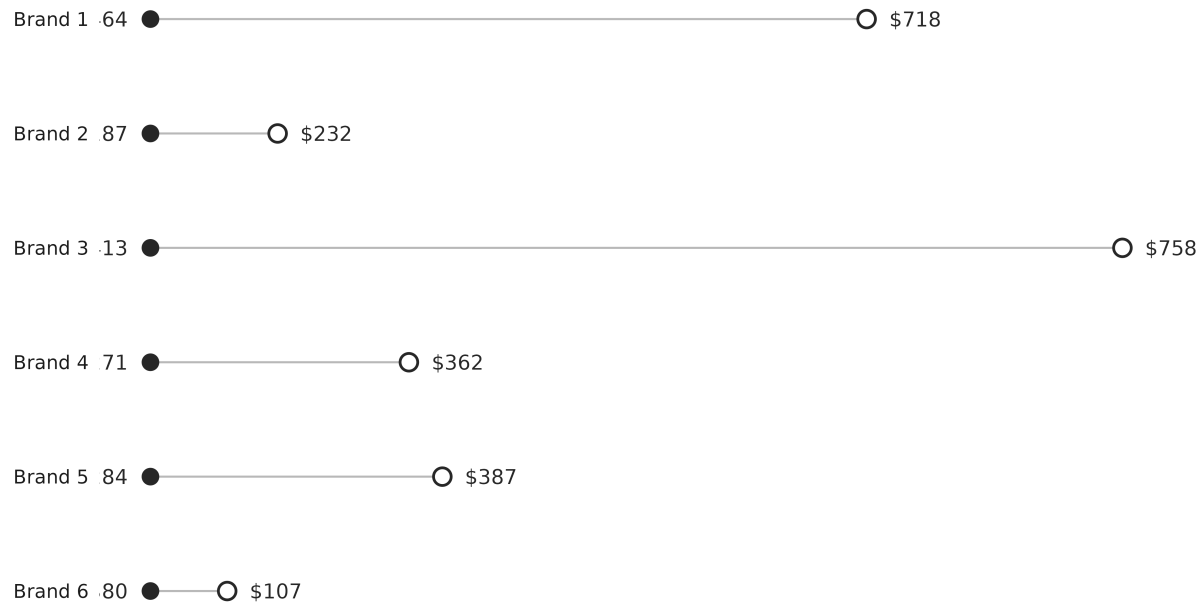
We define targeted brand displacement as the total absolute movement across those five relative brand positions. In these results, direction is not being hidden by the absolute-value summary: all 90 competitor contrasts move in the expected direction. Aggregate displacement is larger under Justified than Base for all six Brand Consideration groups; Justified exceeds Base in 28 of the 30 underlying pairwise contrasts.

Figure 3B shows the comparison between Base and Justified. Targeted brand WTP displacement is larger under Justified than under Base for all six Brand Consideration groups.

Targeted brand WTP displacement

Base aligned for comparison; line length shows the increase under Justified.

● Base ○ Justified



Note. Values are total absolute WTP displacement across the five competing brands.

Within each consideration group, the focal brand is the WTP reference; horizontal distance shows the increase from Base to Ju

Larger WTP differences are not intrinsically more accurate or more desirable. That is not the claim. The question left open by Sections 7 and 9 is whether the favorable transfer result under Justified was achieved simply by suppressing the targeted differentiation that motivated adding Brand Consideration in the first place.

It was not.

In this application, targeted differentiation remains under Justified while new-respondent prediction is also better than under Permissive. Base is the relevant benchmark in Figure 3B because the question is not which specification produces the largest subgroup differences. The question is whether regularization simply returns the model to a state in which the added respondent information contributes little targeted differentiation. It does not.

10.3 Two different evidentiary jobs

The two figures answer different questions and should not be read as parallel model comparisons.

The placebo pricing figure establishes a possibility: unsupported covariate-attributed differentiation can propagate into a decision-relevant output. It does not establish how often that occurs or the typical magnitude of the consequence.

The Brand Consideration figure completes the constructive case for this application. Justified's better new-person prediction relative to Permissive was not purchased by eliminating the targeted differentiation that motivated adding Brand Consideration. The comparison with Base shows that targeted differentiation remains under regularization.

Together, the two figures complete the behavioral and economic test. Differences in covariate attribution can reach quantities used for substantive decisions, while regularization need not achieve better transfer by simply flattening the targeted structure.

11. Discussion

The central result is that a hierarchical model can assign substantial systematic preference structure to respondent information, receive strong in-sample rewards for doing so, and still fail when that structure is asked to transfer to new people.

The Twenty Placebos stress test establishes that this failure can occur. The two applied covariate cases are much less dramatic, but the comparative direction is the same: Justified predicts new respondents better than Permissive in all three cases examined here. Taken together, these results show a failure mode and provide affirmative evidence that covariate-level regularization can materially reduce it in the specifications studied here.

11.1 From inclusion to attribution

Discussions of respondent covariates often begin with a natural question:

Which respondent variables should be included?

The results here suggest a second question:

How much systematic preference differentiation should the model attribute to the respondent information that is included?

These are different questions, even though a statistical procedure may address both. A respondent characteristic need not be either useless or entitled to the ordinary upper-level scale. It may contain some useful information while supporting considerably less systematic differentiation than a Permissive specification can fit.

We use attribution as language for understanding that problem. Attribution is not itself a new statistical method. The statistical intervention examined here is group shrinkage, implemented through the Justified Covariates specification. Other regularization structures could address the same problem.

A fixed tighter prior is one such alternative. The fixed upper-level prior in the bayesm exercise reduced the apparent in-sample advantage produced by twenty random respondent covariates, but did not eliminate it. This shows that fixed shrinkage can materially attenuate the in-sample phenomenon. It also leaves the practical question unresolved. Tightening the prior is mathematically trivial. Choosing how much to tighten it is not.

Justified Covariates address that problem by allowing the amount of covariate-level contraction to be learned within the hierarchy, conditional on a specified hyperprior, rather than fixing the coefficient scale entirely in advance.

The question, then, is not whether shrinkage is available, but what the demonstrated failure and response add to what is already known about respondent covariates.

11.2 What this adds to prior work

Respondent covariates have long been used to explain preference heterogeneity in hierarchical choice models, and both academic and practitioner work has examined how such information should enter the hierarchy and which covariates appear useful. Rossi, McCulloch, and Allenby (1996), Shively, Allenby, and Kohn (2000), and later applied work establish much of that foundation.

More importantly for the present results, Dotson, Büschken, and Allenby (2020) already provide evidence that better in-sample fit from respondent covariates need not translate into better prediction for genuinely held-out consumers. In one application, direct use of observed respondent covariates slightly improves fit while substantially worsening prediction for new respondents.

The present study extends that observation in several concrete ways. The respondent-level mapping between the observed characteristic and preference is deliberately broken through random shuffling, creating a controlled stress test in which the resulting respondent information should not carry transferable preference structure. The resulting Permissive model nevertheless receives an extraordinary in-sample reward. We then inspect where the fitted heterogeneity is placed, regularize the covariate-attributed component, test transfer on untouched respondents, and trace the resulting structure into downstream economic quantities.

That sequence is the addition. It does not require the stronger claim that overfit, shrinkage, or attribution itself is new.

11.3 What the results imply for practice

Adding respondent covariates should not be treated as free model enrichment. If an added variable improves fit, the model has demonstrated that it can use the variable within the estimation sample. It has not demonstrated that the resulting preference structure transfers.

That distinction becomes especially important when comparing specifications. **If the claim is that one model generalizes better than another, that superiority cannot be established using only the data that fitted the models.** Evidence for the superiority claim must come from respondents who's choice data played no role in estimation.

A separate field study is not required to create such evidence. A sample can be split so that some respondents remain completely outside model estimation. Lack of budget for a second data collection therefore does not, by itself, justify making a generalization claim from estimation-sample evidence alone. Other legitimate new-person validation designs are possible as well; the essential requirement is that the

evidence used to establish transfer be external to the fitting of the models being compared.

This point is deliberately demanding. Stronger claims require evidence capable of supporting them. Calling estimation tasks “holdouts,” reporting excellent fit, or showing differentiated subgroup estimates does not turn estimation-sample information into evidence about new consumers.

Base is equally important. A regularized specification has not solved the problem merely because it beats a damaged Permissive model. The relevant comparison also includes what happens when the questionable respondent information is not given the opportunity to generate the additional systematic structure.

11.4 Beyond prediction

Prediction is the central test of transfer in this paper, but transfer alone does not finish the evaluation.

A respondent-covariate model faces two different burdens. Poor-information covariates should not receive consequential systematic attribution merely because the model can fit it. At the same time, a regularization method that improves transfer only by eliminating the targeted differentiation that motivated adding respondent information has achieved a limited kind of success.

The fitted-heterogeneity and downstream analyses therefore are not additional model-comparison contests. They ask what the predictive result means. The decomposition shows where the fitted model places heterogeneity; the behavioral and economic analyses show whether the resulting differentiation reaches quantities the model is intended to inform.

Larger heterogeneity or larger subgroup differences are not inherently desirable. Neither is less differentiation automatically better. The relevant question is whether the systematic differentiation attributed to respondent information is supported by evidence appropriate to what that differentiation is being asked to do.

11.5 Evidence must match the claim

The broader lesson is evidentiary: no single favorable statistic can establish all of the claims a respondent-covariate model might be asked to support.

Likelihood and marginal density address in-sample evidence. New-person prediction addresses transfer. Coherence diagnostics address the competing explanation of obvious estimation failure. The posterior decomposition addresses where the fitted model attributes heterogeneity, and downstream calculations address whether those differences reach substantive quantities.

That sequence is intentional. It follows the MES logic introduced in Section 4: evidence should be matched to the claim it is capable of supporting. This paper does not test MES; it uses that evidentiary logic to organize the evaluation.

No one statistic carries all of those claims.

That principle reaches beyond this particular estimator. Systematic preference differentiation attributed to respondent information should not receive evidentiary credit merely because the model can estimate it. It should earn support from evidence capable of establishing the claim being made.

12. Limitations and Future Research

The evidence in this paper supports a bounded claim. It demonstrates a failure mode, shows how that failure appears inside the fitted hierarchy, and evaluates one regularization response. It does not establish how frequently the problem occurs across hierarchical choice applications or that the Justified specification will always be preferable.

12.1 One application and one stress test

The primary evidence comes from one durable-goods application. The Twenty Placebos condition is one simultaneous stress test involving twenty separately generated placebo permutations. It is not twenty independent replications and is not intended to represent a typical applied specification. Its role is to establish possibility under one deliberately demanding realization, not the frequency or expected magnitude of the failure across applications.

The underlying study was designed and fielded for a substantive commercial decision problem rather than for the covariate hypothesis examined here. Purpose-built studies could manipulate covariate number, strength, prevalence, and alignment more directly.

The severe failure is also not automatic. A single-placebo specification does not reproduce the Twenty Placebos collapse. Separately, the bayesm exercise shows that the apparent in-sample benefit of random upper-level covariates becomes much more pronounced at larger covariate counts, although the pattern is not strictly monotonic. That exercise does not establish an out-of-sample dose-response relationship.

Replication across datasets, sample sizes, model dimensionalities, covariate counts, and prior structures would help establish the boundary conditions under which permissive respondent-covariate modeling becomes consequential. Those are questions of prevalence and scope beyond the possibility result established here.

12.2 Justified Covariates are one regularization structure

The Justified specification examined here uses one shared regularizing scale for the vector of upper-level effects associated with each respondent covariate. That is a deliberate structural choice. A covariate whose relationship with preference is concentrated in only a small subset of coefficients may behave differently under a row-wise group scale.

The results are also conditional on the prior structure examined here. As with any Bayesian model, the amount of regularization depends on that specification. The relevant question for the present paper is whether the empirical conclusions are properties of an obviously knife-edge setting; sensitivity analysis can address that narrower issue without requiring a general theory of prior choice. Doubling the

justification-scale prior from $\sigma_s = 0.5$ to 1 left the central Twenty Placebos result essentially unchanged: new-respondent prediction remained at approximately the Base benchmark and far above the Permissive specification.

Other regularization structures could impose skepticism differently. Spike-and-slab formulations, for example, provide a more explicit route toward inclusion or exclusion. The group-shrinkage approach used here instead allows a respondent characteristic to remain in the hierarchy while regulating the magnitude of the systematic differentiation associated with it. That distinction reflects the modeling problem considered here: a covariate may contain some useful information without warranting the full systematic latitude available under the Permissive specification.

We do not compare alternative regularizers on the primary application. The bayesm exercise examines fixed shrinkage only as an in-sample sensitivity check; it does not compare new-person predictive performance. The current results therefore support Justified Covariates as one examined response to the failure mode, not as the uniquely correct regularization method.

Other approaches may control this failure as well or better. That would be a welcome result; the commitment here is to the modeling problem, not to this particular regularization structure. Improved fit alone would not establish that the resulting structure transfers, nor that its downstream consequences for decision making are benign.

12.3 What makes covariate-driven differentiation transferable?

Feature Importance 8 and Brand Consideration 6 produce different empirical patterns, but the present study cannot tell us why. The two covariate sets differ simultaneously in content, signal strength, prevalence, and substantive proximity to the preferences they are intended to explain. Their comparison therefore does not support a general taxonomy of good and bad covariates.

The comparison does, however, open a more useful research question: What characteristics allow respondent information to support systematic preference differentiation that transfers to new people?

Potential dimensions include signal strength, prevalence, stability across samples, substantive proximity to the preference being modeled, the number of covariates supplied jointly, and whether the relationship is concentrated in a few preference parameters or spread broadly across the vector. These are research questions, not conclusions from the present study.

A related question is when permissive covariate latitude becomes practically consequential and which regularization structures reliably control that risk without suppressing transferable signal. Those questions require additional applications and purpose-built designs. They do not change what the present evidence establishes.

13. Conclusion

Respondent covariates can add useful systematic structure to hierarchical choice models. But inclusion alone does not establish that the preference differentiation

attributed to those covariates will transfer beyond the estimation sample. In this study, permissive treatment produced respondent-conditioned structure that looked extraordinarily successful in sample and then failed when asked to predict new people. Regularizing that structure substantially reduced the failure in the cases examined here.

The implication is not that respondent covariates should be avoided. The amount of systematic preference differentiation assigned to them requires evidence of its own, particularly because that structure can propagate into willingness-to-pay estimates, pricing results, and other quantities used to support decisions. In the Brand Consideration case examined here, regularization improved new-person prediction without eliminating targeted differentiation.

A natural expectation is that irrelevant respondent information will receive little posterior influence. The Twenty Placebos result shows that permissive treatment does not guarantee that outcome. In this application, covariate-level shrinkage comes much closer: the irrelevant covariates remain in the model, but their systematic contribution is strongly contracted and the resulting model returns close to the no-covariate benchmark across the evidence examined here.

The broader principle is earned heterogeneity. Systematic preference differentiation should not receive evidentiary credit simply because the model can estimate it. It should be supported by evidence capable of establishing the claim being made.

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Appendix

Sensitivity to the justification-scale prior

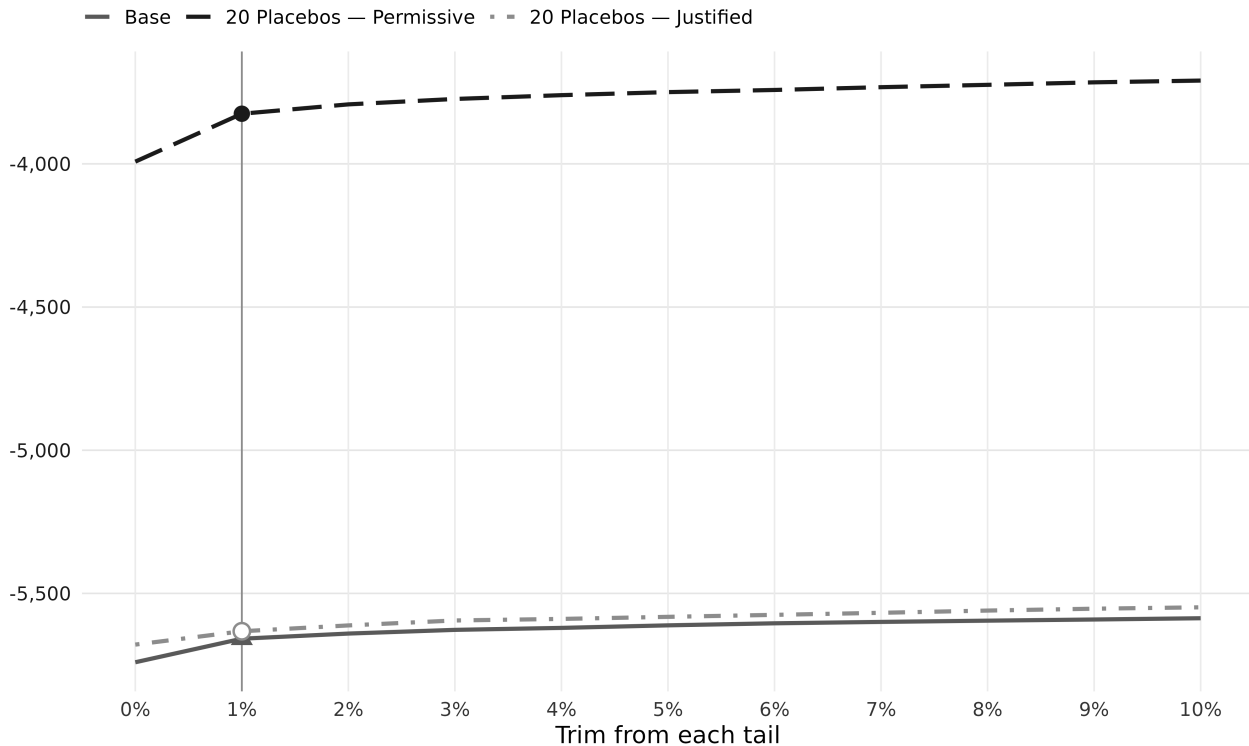
The primary Justified specification uses $\sigma_s = 0.5$. To test whether the Twenty Placebos result depends materially on that choice, we refit the model with $\sigma_s = 1$. The validation result was effectively unchanged. Mean log PPL relative to Base was +0.047 for $\sigma_s = 0.5$ and -0.014 for $\sigma_s = 1$, compared with -9.498 for Permissive. The mean difference between the two Justified specifications was -0.062 log PPL units per validation respondent. In-sample likelihood and NR-LMD likewise remained close to Base under both Justified specifications. The learned justification scales were nearly identical across the two fits.

Sensitivity of the Twenty-Placebo Result to the Justification-Scale Prior			
Specification	Δ mean log likelihood vs. Base	Δ 1%-trimmed NR-LMD vs. Base	Δ mean new-respondent log PPL vs. Base
Justified, $\sigma_s = 0.5$	+43.5	+26.0	+0.047
Justified, $\sigma_s = 1.0$	+42.8	+34.4	-0.014
Permissive	+1907.3	+1833.3	-9.498
<i>Values are differences relative to Base. Higher values are better. NR-LMD uses the 1% trimmed calculation. New-respondent log PPL is reported per validation respondent.</i>			

Trimming NR-LMD

Newton-Raftery Log Marginal Density Across Trimming Levels

Trim from each tail varies from 0% to 10%; the vertical line marks the manuscript's primary 1% specification.



Newton-Raftery log marginal density for Base, 20-placebo Permissive, and 20-placebo Justified specifications as the likelihood.

Independent bayesm exercise: reproducibility and prior sensitivity

The independent bayesm exercise also provides a simple check on two alternative explanations for the in-sample result: dependence on one particular random-covariate realization and dependence on the package's default upper-level prior (Rossi, 2025). We therefore estimated two additional 20-covariate specifications. One retained the original random covariates but replaced the default prior precision, $Ad=0.01 \cdot I$, with a substantially tighter fixed prior, $Ad=I$ (10X SD difference). The other retained the default prior but used a newly generated set of twenty random respondent covariates.

The new random realization reproduced the favorable in-sample comparison. Tightening the fixed prior materially reduced that apparent advantage, but both mean log likelihood and Newton-Raftery log marginal density remained above the no-covariate model. This exercise is entirely in sample. It therefore does not establish which fixed prior predicts better or identify a correct amount of shrinkage; its role is to show that the apparent random-covariate advantage is reproducible and materially sensitive to an analyst-chosen prior scale.

Appendix Table A. Fixed-prior sensitivity in the independent bayesm exercise

20-covariate condition	Upper-level prior	Mean log likelihood		NR-LMD	
		Estimate	Δ vs Base	Estimate	Δ vs Base
No added covariates	—	−2,861.267	0.000	−2,942.818	0.000
Original random realization	Default, $A_d = .011$	−2,723.491	+137.775	−2,804.404	+138.414
Original random realization	Tight, $A_d = 1$	−2,776.695	+84.572	−2,869.027	+73.791
New random realization	Default, $A_d = .011$	−2,729.590	+131.676	−2,837.925	+104.893

Note. Twenty centered random respondent covariates enter the upper-level model for all ten preference parameters. The tight-prior condition changes only the prior precision on the upper-level covariate coefficients. This exercise evaluates in-sample evidence only and does not compare new-person prediction.

Note. The default bayesm specification uses for the upper-level covariate coefficients; the tight-prior condition uses . Choice data, twenty centered random respondent covariates, heterogeneity specification, and MCMC settings are otherwise held fixed for the prior comparison. NR-LMD uses the same trimming convention as the primary bayesm exercise. No new-person validation is performed in this exercise.

Holdout Predictions: PPL

Here y_i^{val} is the complete validation response vector, D_{est} is the estimation sample, z_i contains the validation respondent's covariates, β_i is the new respondent's preference vector, and Θ contains the fitted population parameters. We approximate the posterior predictive probability by Monte Carlo averaging over posterior draws and corresponding draws from the new-respondent heterogeneity distribution, following Gelman, Meng, and Stern (1996). Respondent-level log PPL is the log of this average.

$$p(y_i^{\text{val}} | z_i, D_{\text{est}}) = \int \left[\int p(y_i^{\text{val}} | \beta_i, \Theta) p(\beta_i | z_i, \Theta) d\beta_i \right] p(\Theta | D_{\text{est}}) d\Theta$$

$$\approx \frac{1}{M} \sum_{d=1}^M p(y_i^{\text{val}} | \beta_i^{(d)}, \Theta^{(d)}), \beta_i^{(d)} \sim p(\beta_i | z_i, \Theta^{(d)})$$

For each Monte Carlo draw, the same $\beta_i^{(d)}$ is used across all validation choice tasks for respondent i .

Coherence Diagnostics Comparison

The residual-scale parameters show some evidence of imperfect mixing. Mildly elevated $\hat{R}$ values occur more frequently in the Twenty Placebos Permissive fit than in the Base or Justified fits, although the pattern is neither unique to Permissive nor uniformly severe. The largest $\hat{R}$ in this block occurs in Base ($\tau_{17} = 1.0588$), and isolated elevations also appear under Justified. These diagnostics therefore do not support describing the fitted models as uniformly pristine. At the same time, they do not identify an obvious convergence failure specific to the Permissive model that would by itself explain its severe new-respondent predictive deterioration.

Gamma Internal Coherence Diagnostics

N = 321 estimation-sample fits

Parameter	Base			20 Placebos — Permissive			20 Placebos — Justified		
	R̂	Bulk ESS	Tail ESS	R̂	Bulk ESS	Tail ESS	R̂	Bulk ESS	Tail ESS
Gamma[1,1]	1.0079	437	707	1.0074	662	890	1.0033	595	770
Gamma[1,2]	1.0002	1,208	1,333	0.9996	1,346	1,416	1.0003	1,648	1,489
Gamma[1,3]	1.0011	1,290	1,281	1.0013	1,471	1,500	1.0024	1,880	1,400
Gamma[1,4]	1.0002	1,187	1,520	1.0078	1,592	1,360	1.0005	1,585	1,500
Gamma[1,5]	1.0017	1,080	1,265	1.0033	1,241	1,594	1.0001	1,590	1,590
Gamma[1,6]	1.0010	1,652	1,420	1.0015	1,454	1,413	1.0033	1,630	1,290
Gamma[1,7]	1.0015	1,095	1,308	1.0021	770	1,345	1.0002	1,590	1,490
Gamma[1,8]	1.0000	1,777	1,307	1.0030	1,513	991	1.0025	2,243	1,720
Gamma[1,9]	1.0023	1,280	1,321	1.0067	1,172	1,019	0.9998	1,805	1,650
Gamma[1,10]	0.9994	1,284	1,203	1.0016	1,184	1,487	1.0006	1,683	1,610
Gamma[1,11]	1.0006	1,657	1,615	1.0031	1,018	1,448	1.0016	1,484	1,430
Gamma[1,12]	1.0011	1,408	1,438	1.0064	1,283	1,528	1.0020	2,134	1,550
Gamma[1,13]	1.0008	1,236	1,490	1.0058	969	1,030	1.0004	2,195	1,460
Gamma[1,14]	1.0010	1,303	1,198	1.0061	742	1,332	1.0026	1,862	1,470
Gamma[1,15]	1.0008	1,693	1,682	1.0023	1,179	1,286	0.9999	2,047	1,550
Gamma[1,16]	1.0019	1,670	1,463	1.0020	1,247	1,504	1.0005	2,572	1,710
Gamma[1,17]	1.0022	1,839	1,566	1.0016	1,456	1,312	1.0022	2,107	1,520
Gamma[1,18]	1.0002	1,522	1,513	1.0038	909	1,474	0.9991	1,802	1,710
Gamma[1,19]	1.0013	1,563	1,392	1.0010	1,338	1,481	1.0018	2,185	1,500
Gamma[1,20]	1.0034	1,624	1,091	1.0074	851	1,173	0.9994	2,271	1,480
Gamma[1,21]	1.0016	1,291	1,293	1.0038	1,386	1,401	1.0025	1,605	1,530
Gamma[1,22]	1.0034	1,617	1,378	1.0033	1,586	1,428	0.9993	2,181	1,740
Gamma[1,23]	1.0019	1,238	1,455	1.0026	1,408	1,390	1.0002	1,484	1,580
Gamma[1,24]	0.9997	1,463	1,545	1.0033	986	1,523	1.0003	1,724	1,440
Gamma[1,25]	1.0001	1,571	1,382	1.0069	726	1,434	0.9998	2,109	1,710
Gamma[1,26]	1.0012	927	1,236	1.0008	1,083	1,579	0.9998	1,571	1,660
Gamma[1,27]	1.0051	1,157	1,267	1.0052	1,412	1,616	0.9992	2,096	1,550
Gamma[1,28]	1.0018	1,392	1,593	1.0047	1,428	1,619	1.0003	1,972	1,520
Gamma[1,29]	1.0040	748	1,104	1.0160	330	842	1.0023	1,212	1,570

Gamma Internal Coherence Diagnostics

N = 321 estimation-sample fits

Parameter	Base			20 Placebos — Permissive			20 Placebos — Justified		
	$\hat{R}$	Bulk ESS	Tail ESS	$\hat{R}$	Bulk ESS	Tail ESS	$\hat{R}$	Bulk ESS	Tail ESS
Gamma[1,1]	1.0079	437	707	1.0074	662	890	1.0033	595	777
Gamma[1,2]	1.0002	1,208	1,333	0.9996	1,346	1,416	1.0003	1,648	1,489
Gamma[1,3]	1.0011	1,290	1,281	1.0013	1,471	1,500	1.0024	1,880	1,400
Gamma[1,4]	1.0002	1,187	1,520	1.0078	1,592	1,360	1.0005	1,585	1,500
Gamma[1,5]	1.0017	1,080	1,265	1.0033	1,241	1,594	1.0001	1,590	1,590
Gamma[1,6]	1.0010	1,652	1,420	1.0015	1,454	1,413	1.0033	1,630	1,290
Gamma[1,7]	1.0015	1,095	1,308	1.0021	770	1,345	1.0002	1,590	1,490
Gamma[1,8]	1.0000	1,777	1,307	1.0030	1,513	991	1.0025	2,243	1,720
Gamma[1,9]	1.0023	1,280	1,321	1.0067	1,172	1,019	0.9998	1,805	1,650
Gamma[1,10]	0.9994	1,284	1,203	1.0016	1,184	1,487	1.0006	1,683	1,610
Gamma[1,11]	1.0006	1,657	1,615	1.0031	1,018	1,448	1.0016	1,484	1,430
Gamma[1,12]	1.0011	1,408	1,438	1.0064	1,283	1,528	1.0020	2,134	1,550
Gamma[1,13]	1.0008	1,236	1,490	1.0058	969	1,030	1.0004	2,195	1,460
Gamma[1,14]	1.0010	1,303	1,198	1.0061	742	1,332	1.0026	1,862	1,470
Gamma[1,15]	1.0008	1,693	1,682	1.0023	1,179	1,286	0.9999	2,047	1,550
Gamma[1,16]	1.0019	1,670	1,463	1.0020	1,247	1,504	1.0005	2,572	1,710
Gamma[1,17]	1.0022	1,839	1,566	1.0016	1,456	1,312	1.0022	2,107	1,520
Gamma[1,18]	1.0002	1,522	1,513	1.0038	909	1,474	0.9991	1,802	1,710
Gamma[1,19]	1.0013	1,563	1,392	1.0010	1,338	1,481	1.0018	2,185	1,500
Gamma[1,20]	1.0034	1,624	1,091	1.0074	851	1,173	0.9994	2,271	1,480
Gamma[1,21]	1.0016	1,291	1,293	1.0038	1,386	1,401	1.0025	1,605	1,530
Gamma[1,22]	1.0034	1,617	1,378	1.0033	1,586	1,428	0.9993	2,181	1,740
Gamma[1,23]	1.0019	1,238	1,455	1.0026	1,408	1,390	1.0002	1,484	1,580
Gamma[1,24]	0.9997	1,463	1,545	1.0033	986	1,523	1.0003	1,724	1,440
Gamma[1,25]	1.0001	1,571	1,382	1.0069	726	1,434	0.9998	2,109	1,710
Gamma[1,26]	1.0012	927	1,236	1.0008	1,083	1,579	0.9998	1,571	1,660
Gamma[1,27]	1.0051	1,157	1,267	1.0052	1,412	1,616	0.9992	2,096	1,550
Gamma[1,28]	1.0018	1,392	1,593	1.0047	1,428	1,619	1.0003	1,972	1,520
Gamma[1,29]	1.0040	748	1,104	1.0160	330	842	1.0023	1,212	1,570

s Internal Coherence Diagnostics

N = 321 estimation-sample fits

Parameter	Base			20 Placebos — Permissive			20 Placebos — Justified		
	$\hat{R}$	Bulk ESS	Tail ESS	$\hat{R}$	Bulk ESS	Tail ESS	$\hat{R}$	Bulk ESS	Tail ESS
s[1]	—	—	—	—	—	—	1.0090	869	981
s[2]	—	—	—	—	—	—	1.0046	539	582
s[3]	—	—	—	—	—	—	1.0004	838	859
s[4]	—	—	—	—	—	—	1.0013	1,002	1,223
s[5]	—	—	—	—	—	—	1.0031	1,016	1,095
s[6]	—	—	—	—	—	—	1.0005	916	1,222
s[7]	—	—	—	—	—	—	1.0021	696	1,018
s[8]	—	—	—	—	—	—	1.0029	749	1,105
s[9]	—	—	—	—	—	—	1.0025	690	1,037
s[10]	—	—	—	—	—	—	1.0004	874	1,201
s[11]	—	—	—	—	—	—	1.0054	473	827
s[12]	—	—	—	—	—	—	1.0069	754	1,449
s[13]	—	—	—	—	—	—	1.0086	492	833
s[14]	—	—	—	—	—	—	1.0005	484	839
s[15]	—	—	—	—	—	—	1.0006	842	674
s[16]	—	—	—	—	—	—	1.0118	621	1,226
s[17]	—	—	—	—	—	—	1.0002	1,419	1,258
s[18]	—	—	—	—	—	—	1.0029	889	1,127
s[19]	—	—	—	—	—	—	1.0017	1,096	980
s[20]	—	—	—	—	—	—	1.0030	645	667

Rows labeled $s[j]$ correspond to the Stan parameter $trust[j]$ in the Justified specification.